

3)

3.1) mínimo $\rightarrow 0,01 \text{ €}$

máximo $\rightarrow 99999999,99 \text{ €}$

3.2) a - $131 : 2 = 65 \cdot 2 + \textcircled{1}$

$65 : 2 = 32 \cdot 2 + \textcircled{1}$

$32 : 2 = 16 \cdot 2 + \textcircled{0}$

$16 : 2 = 8 \cdot 2 + \textcircled{0}$

$8 : 2 = 4 \cdot 2 + \textcircled{0}$

$4 : 2 = 2 \cdot 2 + \textcircled{0}$

$2 : 2 = 1 \cdot 2 + \textcircled{0}$

$1 : 2 = 0 \cdot 2 + \textcircled{1}$

$131 = 10000011$

b - $511 : 2 = 255 \cdot 2 + 1$

$255 : 2 = 127 \cdot 2 + 1$

$127 : 2 = 63 \cdot 2 + 1$

$63 : 2 = 31 \cdot 2 + 1$

$31 : 2 = 15 \cdot 2 + 1$

$15 : 2 = 7 \cdot 2 + 1$

$7 : 2 = 3 \cdot 2 + 1$

$3 : 2 = 1 \cdot 2 + 1$

$1 : 2 = 0 \cdot 2 + 1$

$511 = 11111111$



$$c - 888 : 2 = 444 \cdot 2 + 0$$

$$444 : 2 = 222 \cdot 2 + 0$$

$$222 : 2 = 111 \cdot 2 + 0$$

$$111 : 2 = 55 \cdot 2 + 1$$

$$55 : 2 = 27 \cdot 2 + 1$$

$$27 : 2 = 13 \cdot 2 + 1$$

$$13 : 2 = 6 \cdot 2 + 1$$

$$6 : 2 = 3 \cdot 2 + 0$$

$$3 : 2 = 1 \cdot 2 + 1$$

$$1 : 2 = 0 \cdot 2 + 1$$

$$888 \rightarrow 1101111000$$

$$d - 4096 : 2 = 2048 \cdot 2 + 0$$

$$2048 : 2 = 1024 \cdot 2 + 0$$

$$1024 : 2 = 512 \cdot 2 + 0$$

$$512 : 2 = 256 \cdot 2 + 0$$

$$256 : 2 = 128 \cdot 2 + 0$$

$$128 : 2 = 64 \cdot 2 + 0$$

$$64 : 2 = 32 \cdot 2 + 0$$

$$32 : 2 = 16 \cdot 2 + 0$$

$$8 : 2 = 4 \cdot 2 + 0$$

$$4 : 2 = 2 \cdot 2 + 0$$

$$2 : 2 = 1 \cdot 2 + 0$$

$$1 : 2 = 0 \cdot 2 + 1$$

$$4096 = 100000000000$$

3.3) a - $2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 31$

b - $2^9 + 2^8 + 2^7 + \dots + 2^0 = 1023$

c - $2^{17} + 2^{16} + \dots + 2^0 = 262143$

d - $2^{31} + 2^{30} + \dots + 2^0 = 4294967295$

3.4) ① Descobrir os dígitos da base 13

↳ cada base usa os dígitos de 0 até $b-1$

Logo 0 até $13-1=12$

↳ Como apenas temos algarismos de 0 ao 9, usamos as letras quando ultrapassamos os 10 números

↳ Usamos, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C

② Pensar cada dígito a binário

Como o dígito mais alto é o C (que vale 12), precisamos de usar sequências de 4 bits (3 bits apenas dá $2^3=8$ combinações (0-7))

0	=	0000
1	=	0001
2	=	0010
3	=	0011
4	=	0100
5	=	0101
6	=	0110
7	=	0111
8	=	1000
9	=	1001
A (10)	=	1010
B (11)	=	1011

$$C(12) = 1100$$

3.5) a- Remove de uma sequência de 4 bits

$$\underline{1011}$$

$$11$$

$$11$$

$$6$$

$$\underline{1110}$$

$$14$$

$$11$$

$$E$$

$$\underline{1101}$$

$$13$$

$$11$$

$$D$$

$$BED_{16}$$

$$b- \underline{10} \quad \underline{0111} \quad \underline{0110}$$

$$2$$

$$7$$

$$6$$

$$276_{16}$$

$$c- \underline{111} \quad \underline{1111} \quad \underline{1111}$$

$$7$$

$$15$$

$$11$$

$$F$$

$$15$$

$$11$$

$$F$$

$$7FF_{16}$$

$$d- \underline{101} \quad \underline{0001} \quad \underline{1110}$$

$$5$$

$$1$$

$$E$$

$$51E_{16}$$

$$3.6) \quad B(11) = 1011 \quad ; \quad E(14) = 1110$$

$$11:2 = 5 \cdot 2 + 1$$

$$14:2 = 7 \cdot 2 + 0$$

$$5:2 = 2 \cdot 2 + 1$$

$$7:2 = 3 \cdot 2 + 1$$

$$2:2 = 2 \cdot 1 + 0$$

$$3:2 = 1 \cdot 2 + 1$$

$$1:2 = 0 \cdot 2 + 1$$

$$1:2 = 0 \cdot 2 + 1$$

$$F(15) = 1111$$

$$BEEF = 1011\ 1110\ 1110\ 1111$$

$$h- \quad 1 = 0001$$

$$0 = 0000$$

$$\text{Logo} \quad 1000 = 0001\ 0000\ 0000\ 0000$$

$$F(15)$$

$$15 : 2 = 7 \cdot 2 + \textcircled{1}$$

$$7 : 2 = 3 \cdot 2 + \textcircled{1}$$

$$3 : 2 = 1 \cdot 2 + \textcircled{1}$$

$$1 : 2 = 0 \cdot 2 + \textcircled{1}$$

$$15 = 1111$$

$$1000.FF = 0001\ 0000\ 0000\ 0000\ 0000.11111111_2$$

$$C- \quad A(10) = 1010$$

$$B(11) = 1011$$

$$C(12) = 1100$$

$$D(13) = 1101$$

$$E(14) = 1110$$

$$F(15) = 1111$$

$$1010\ 1011\ 1100.1101\ 1110\ 1111$$

$$d - 110110101100.00110100$$

$$3.7) a - 77 = 5 \cdot 15 + 2$$

$$15 : 2 = 3 \cdot 5 + 0$$

$$3 : 5 = 0 \cdot 5 + 3$$

$$302_5$$

$$b - 131 : 5 = 26 \cdot 5 + 1$$

$$26 : 5 = 5 \cdot 5 + 1$$

$$5 : 5 = 1 \cdot 5 + 0$$

$$1 : 5 = 0 \cdot 5 + 1$$

$$1011_5$$

$$c - 511 : 5 = 5 \cdot 102 + 1$$

$$102 : 5 = 5 \cdot 20 + 2$$

$$20 : 5 = 5 \cdot 4 + 0$$

$$4 : 5 = 0 \cdot 5 + 4$$

$$4021_5$$

$$d - 1000 : 5 = 200 \cdot 5 + 0$$

$$200 : 5 = 40 \cdot 5 + 0$$

$$40 : 5 = 8 \cdot 5 + 0$$

$$8 : 5 = 1 \cdot 5 + 3$$

$$1 : 5 = 0 \cdot 5 + 1$$

$$13000_5$$

$$3.8) a - 6 \cdot 9^1 + 6 \cdot 9^0 = 54 + 6 = 60$$

$$b - 1 \cdot 9^2 + 2 \cdot 9^1 + 3 \cdot 9^0 = 102$$

$$c - 3 \cdot 9^2 + 1 \cdot 9^1 + 7 \cdot 9^0 = 259$$

$$d - 8 \cdot 9^2 + 0 + 0 = 648$$

$$3.9) a - 66 : 2 = 33 \cdot 2 + 0$$

$$33 : 2 = 16 \cdot 2 + 1$$

$$16 : 2 = 8 \cdot 2 + 0$$

$$8 : 2 = 4 \cdot 2 + 0$$

$$2 : 2 = 1 \cdot 2 + 0$$

$$1 : 2 = 0 \cdot 2 + 1$$

$$66_{10} = 100010_2$$

$$66 : 7 = 9 \cdot 7 + 3$$

$$9 : 7 = 1 \cdot 7 + 2$$

$$1 : 7 = 0 \cdot 7 + 1$$

$$66_{10} = 123_7$$

$$66 : 9 = 7 \cdot 9 + 3$$

$$7 : 9 = 0 \cdot 9 + 7$$

$$66_{10} = 73_9$$

$$\begin{aligned} b - 13F.4_{16} &= 16^2 + 3 \cdot 16^1 + 13 \cdot 16^0 + 4 \cdot 16^{-1} \\ &= 256 + 48 + 13 + 0,25 \\ &= 319,25_{10} \end{aligned}$$

$$319 : 12 = 26 \cdot 12 + 7$$

$$26 : 12 = 2 \cdot 12 + 2$$

$$2 : 12 = 0 \cdot 12 + 2$$

$$0,25 \cdot 12 = 3,0$$

$$13F,4_{16} = 227,3_{12}$$

$$\begin{aligned} \text{C- } 1110010,1 &= 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^1 + 1 \cdot 2^{-1} \\ &= 114,5_{10} \end{aligned}$$

$$114 : 3 = 38 \cdot 3 + 0$$

$$38 : 3 = 12 \cdot 3 + 2$$

$$12 : 3 = 4 \cdot 3 + 0$$

$$4 : 3 = 1 \cdot 3 + 1$$

$$1 : 3 = 0 \cdot 3 + 1$$

$$0,5 \cdot 3 = 1,5$$

↳ vai dar sempre 1,5

logo a parte inteira e' sempre 1

$$1110010,1_2 = 11020,(1)_3$$

$$114 : 4 = 28 \cdot 4 + 2$$

$$28 : 4 = 7 \cdot 4 + 0$$

$$7 : 4 = 1 \cdot 4 + 3$$

$$1 : 4 = 0 \cdot 4 + 1$$

$$0,5 \cdot 4 = 2$$

$$1302 \cdot 2_4$$

$$114 : 7 = 16 \cdot 7 + 2$$

$$16 : 7 = 2 \cdot 7 + 2$$

$$2 : 7 = 0 \cdot 7 + 2$$

$$0,5 \cdot 7 = 3,5$$

Logo a parte inteira vai reter 3

$$222, (3)_7$$

d- $A = 10$
 $B = 11$

$$\begin{aligned} 4B7_{13} &= 10 \cdot 13^2 + 11 \cdot 13 + 7 \cdot 13^0 \\ &= 1690 + 143 + 7 \\ &= 1840_{10} \end{aligned}$$

$$1840 : 2 = 920 \cdot 2 + 0$$

$$920 : 2 = 460 \cdot 2 + 0$$

$$460 : 2 = 230 \cdot 2 + 0$$

$$230 : 2 = 115 \cdot 2 + 0$$

$$115 : 2 = 57 \cdot 2 + 1$$

$$57 : 2 = 28 \cdot 2 + 1$$

$$28 : 2 = 14 \cdot 2 + 0$$

$$14 : 2 = 7 \cdot 2 + 0$$

$$7 : 2 = 2 \cdot 2 + 3$$

$$3 : 2 = 1 \cdot 2 + 1$$

$$1 : 2 = 0 \cdot 2 + 1$$

$$AB7_{13} = 11100110000_2$$

$$1840 : 6 = 306 \cdot 6 + 4$$

$$306 : 6 = 51 \cdot 6 + 0$$

$$51 : 6 = 8 \cdot 6 + 3$$

$$8 : 6 = 1 \cdot 6 + 2$$

$$1 : 6 = 0 \cdot 6 + 1$$

$$AB7_{13} = 12304_6$$

$$1840 : 8 = 230 \cdot 8 + 0$$

$$230 : 8 = 28 \cdot 8 + 6$$

$$28 : 8 = 3 \cdot 8 + 4$$

$$3 : 8 = 0 \cdot 8 + 3$$

$$AB7_{13} = 3460_8$$

3.10) 1ª Linha (horas)

↳ usa sistema de 5 bits

$$H = 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 \\ = 9$$

2ª Linha (minutos)

↳ 6 bits

$$M = 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 \\ = 25$$

3ª Linha (segunda)

↳ 6 bits

$$S = 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 \\ = 46$$

As horas são 9:25:46